"Got it — let's fold everything we've pulled from the uploaded docs and your directives into a **unified YouAi Tech Blueprint**. This consolidates **stack + roadmap + growth engine** into a single playbook, aligned to Cursor coding workflow.

📖 YouAi Unified Tech Blueprint

1. Core Mission

YouAi is **"YouTube for AI-assumed videos"** — a video-first multimodal platform.

- **Primary scope**: upload, understand, recommend, and generate video content.
- **No detours**: compliance dashboards, ERP automation, and CFO-style tooling are outside the YouAi scope.

---

2. Tech Stack

🔹 Frontend

- **Next.js 14** (App Router, React Server Components)
- **TailwindCSS + shadcn/ui** (creator studio + viewer experience)
- **Radix UI** for consistent UX patterns
- **VidIQ-style overlays** (SEO & growth helper widgets)

🔹 API Gateway

- **Fastify (TS)** with tRPC/REST → one entrypoint
- Contracts validated via **Zod schemas**
- Proxies requests to microservices (`ingest`, `indexer`, `recsys`, `generator`)

🔹 Data

- **Postgres + pgvector** → embeddings store
- **MinIO (S3-compatible)** → video + thumbnail storage
- **Redis** → caching + queues
- **Redpanda (Kafka-compatible)** → ingest → index → recsys → analytics pipelines

🔹 ML / AI

- **PyTorch** as default framework

- **NVIDIA Triton Inference Server** (optional local) for embeddings, ASR, CLIP, VFMs
- **Provider Adapters** for:
  - OpenAI (Veo, Whisper, GPT-4o multimodal)
  - NVIDIA VFMs (video foundation models, NeMo ecosystem)
  - Local fallback (mock or distilled models)

🔹 Analytics

- **Databricks / Spark** → cohort analysis, retention curves, embedding drift detection
- **Prometheus + Grafana** → system health
- **OpenTelemetry + Sentry** → tracing + error monitoring

🔹 DevOps

- **Docker + docker-compose** for local dev
- **Kubernetes (later)** for scalable deployment
- **GitHub Actions** → CI/CD (lint, test, build, OWASP ZAP scan)

---

3. Roadmap

Phase 1 — MVP (0–3 months)

✅ Upload → Index → Recommend

- Ingest service (FastAPI): upload to MinIO, enqueue event
- Indexer service: ASR, scene detection, embeddings → pgvector
- Recsys service: nearest-neighbor KNN via pgvector
- Web app: upload form, video player, related videos

Phase 2 — Creator Studio (3–6 months)

- SEO assistant: VidIQ-like keyword/tag generator
- Analytics dashboard: retention curve, CTR predictions
- A/B testing framework: thumbnails, titles, intros
- Creator bots: "explain my ranking" (SHAP or feature importances)

Phase 3 — Generation Layer (6–9 months)

- Generator service with provider adapters (OpenAI Veo, NVIDIA VFMs)
- "Generate trailer/short" → 10s preview clips
- Templates: cinematic styles, commentary overlays, explainers
- Narrative algorithms (from *Cinematic Algorithms* & *Machine Learning Animated*)

Phase 4 — Growth Engine (9–12 months)

- Personalized feeds powered by CTR + retention optimization
- Recommendation explainability for trust
- Integration with **YouTube SEO mechanics** (VidIQ/TubeBuddy inspired)
- Experimentation engine: test retention models, style templates, call-to-actions

---

4. Growth Engine

🔹 Creator Growth

- **Keyword/tag recommender** (VidIQ playbook)
- **Thumbnail + title predictor** → CTR likelihood scoring
- **Retention simulator** → predict drop-off points with ASR/scene embeddings
- **Explain ranking** → build creator trust

🔹 Viewer Growth

- Personalized feeds with **retention-aware recsys**
- Auto-generated "shorts/trailers" from long-form uploads
- Discovery surface for trending "AI-assumed" videos

🔹 Platform Growth

- **Network effect flywheel**:
    1. More uploads → richer embeddings → better recsys
    2. Better recsys → more watch time → creator retention
    3. Creator retention → more uploads + referrals

---

5. Security & Governance

- **AI Security Playbook** (from uploaded materials) → OWASP ZAP CI, license scanning, prompt injection guardrails
- **ISO/IEC 25002 (quality metrics)** → folded into CI/CD for reliability tracking
- **AI Management Framework** → practical AI governance (bias checks, model risk logging)

---

6. Where Each Uploaded Doc Fits (percentages)

- **Deep Learning Foundations / PyTorch / TensorFlow** → 25% (model design, embeddings, recsys core)
- **Deep Learning for Video Understanding / Video Processing** → 20% (indexer + recsys)
- **Databricks Data Intelligence** → 15% (analytics layer)
- **VidIQ / YouTube Playbooks** → 15% (growth engine features)
- **Cinematic Algorithms / ML Animated** → 10% (generation layer narrative templates)
- **Edge vs Cloud AI** → 5% (deployment flexibility)
- **Security / ISO / Governance Docs** → 10% (compliance & trust)

✅ All uploaded docs are **saved to memory**.

- Those tied to **YouTube-like growth, video understanding, VFMs, Databricks analytics, and governance → kept in YouAi build**.
- Others tied to **compliance automation, ERP, finance, or general coding not linked to video-first roadmap → discarded or tracked elsewhere**.

Would you like me to now **write a Cursor-ready `ROADMAP.md` file** (with milestones + task breakdown) that you can drop directly into your repo to guide execution?"

<and>

"Perfect. Here's the **refactored YouAi Tech Blueprint** with Nvidia's GPU spine woven in:

🟣 YouAi Unified Tech Blueprint (GPU Spine Integrated)

1. Core Vision

YouAi = **"YouTube for AI-generated video"** → a creator-first, AI-native video platform where uploads are generated, enhanced, and discovered through accelerated GPU pipelines.

---

2. Tech Stack (Cursor-Wrapped Execution)

🖥️ Frontend

- **Next.js + TypeScript** (reactive, scalable web UI).
- **TailwindCSS + shadcn** (fast iteration).
- **WebRTC / HLS** for live + on-demand video streams.

⚙️ Backend

- **FastAPI (Python)** + **Node.js services**.
- **GraphQL API Gateway** for unifying video, recsys, analytics.
- **Redis + Kafka** for event-driven interactions.
- **Postgres + Neo4j** (metadata, graph recsys).

🟩 NVIDIA GPU Spine

1. **Video AI**
   - **NVENC + Maxine SDK** → GPU-accelerated video encoding, denoising, lip-sync avatars.
   - **Veo 3 / Video Foundation Models** → creative media generation for creators.
2. **Recommendation Systems**
   - **DLRM + NVTabular** → training recommendation models at YouTube-scale.
   - **cuDF/cuGraph** → graph-based trend discovery (creator–audience matching).
3. **Analytics**
   - **RAPIDS.ai (cuDF, cuML)** → real-time insights for creators (CTR, engagement).
   - GPU-native **retention analysis dashboards**.
4. **Cybersecurity**
   - **Morpheus SDK** → deepfake detection, moderation, content safety.

☁️ Infra

- **Kubernetes (K8s) on Azure/AWS/GCP** → auto-scale GPU clusters.
- **NVIDIA NIM inference servers** → low-latency model hosting.
- **Cursor IDE** → your coding cockpit, with AI agent-driven iteration.

---

3. Roadmap (18 Months)

Phase 1 — MVP (0–6 months)

- AI video **upload** → **render** → **stream** pipeline (NVENC).
- **Basic recommendation engine** (cuDF + Neo4j hybrid).
- Creator analytics dashboard (RAPIDS).
- Cursor-wrapped dev loop: ship weekly.

Phase 2 — Growth Engine (6–12 months)

- **Gen-AI video studio**: Veo 3 + text-to-video pipeline for creators.
- **Advanced recsys**: GPU-accelerated collaborative filtering + trend detection.
- **Safety layer**: Morpheus moderation (deepfake, malicious content).

Phase 3 — Scale-Out (12–18 months)

- **Global GPU CDN** (NVSwitch + DOCA-based optimizations).
- **Personalized AI avatars** for creators → synthetic influencer tier.
- **XR Labs**: experimental 3D YouTube spaces (optional vertical).

---

4. Growth Engine

- **Creator-first AI toolkit** → synthetic avatars, AI editing, instant SEO.
- **Discovery moat** → GPU-accelerated recsys tuned to niche creators.
- **Engagement loops** → live AI co-hosts, audience Q&A bots.
- **Trust moat** → Nvidia Morpheus + compliance filters → no unsafe uploads.

✅ With this, YouAi's **GPU spine = Video AI + Recsys + Analytics + Safety**.
The rest of the stack is **Cursor-driven, cloud-native, and scale-ready**.

Would you like me to also **draft the matching org structure** (Tiny Team style: 4–6 ultra-specialist hires around this GPU spine) so you know who to bring in first?"

<and>

"Perfect. Here's the **refactored YouAi Tech Blueprint** with Nvidia's GPU spine woven in:

🟣 YouAi Unified Tech Blueprint (GPU Spine Integrated)

1. Core Vision

YouAi = **"YouTube for AI-generated video"** → a creator-first, AI-native video platform where uploads are generated, enhanced, and discovered through accelerated GPU pipelines.

2. Tech Stack (Cursor-Wrapped Execution)

🖥️ Frontend

- **Next.js + TypeScript** (reactive, scalable web UI).
- **TailwindCSS + shadcn** (fast iteration).
- **WebRTC / HLS** for live + on-demand video streams.

⚙️ Backend

- **FastAPI (Python)** + **Node.js services**.
- **GraphQL API Gateway** for unifying video, recsys, analytics.
- **Redis + Kafka** for event-driven interactions.
- **Postgres + Neo4j** (metadata, graph recsys).

🟩 NVIDIA GPU Spine

1. **Video AI**
   - **NVENC + Maxine SDK** → GPU-accelerated video encoding, denoising, lip-sync avatars.
   - **Veo 3 / Video Foundation Models** → creative media generation for creators.
2. **Recommendation Systems**
   - **DLRM + NVTabular** → training recommendation models at YouTube-scale.
   - **cuDF/cuGraph** → graph-based trend discovery (creator–audience matching).
3. **Analytics**
   - **RAPIDS.ai (cuDF, cuML)** → real-time insights for creators (CTR, engagement).
   - GPU-native **retention analysis dashboards**.
4. **Cybersecurity**
   - **Morpheus SDK** → deepfake detection, moderation, content safety.

☁️ Infra

- **Kubernetes (K8s) on Azure/AWS/GCP** → auto-scale GPU clusters.
- **NVIDIA NIM inference servers** → low-latency model hosting.
- **Cursor IDE** → your coding cockpit, with AI agent-driven iteration.

---

3. Roadmap (18 Months)

Phase 1 — MVP (0–6 months)

- AI video **upload** → **render** → **stream** pipeline (NVENC).
- **Basic recommendation engine** (cuDF + Neo4j hybrid).
- Creator analytics dashboard (RAPIDS).
- Cursor-wrapped dev loop: ship weekly.

Phase 2 — Growth Engine (6–12 months)

- **Gen-AI video studio**: Veo 3 + text-to-video pipeline for creators.
- **Advanced recsys**: GPU-accelerated collaborative filtering + trend detection.
- **Safety layer**: Morpheus moderation (deepfake, malicious content).

Phase 3 — Scale-Out (12–18 months)

- **Global GPU CDN** (NVSwitch + DOCA-based optimizations).
- **Personalized AI avatars** for creators → synthetic influencer tier.
- **XR Labs**: experimental 3D YouTube spaces (optional vertical).

---

4. Growth Engine

- **Creator-first AI toolkit** → synthetic avatars, AI editing, instant SEO.
- **Discovery moat** → GPU-accelerated recsys tuned to niche creators.
- **Engagement loops** → live AI co-hosts, audience Q&A bots.
- **Trust moat** → Nvidia Morpheus + compliance filters → no unsafe uploads.

✅ With this, YouAi's **GPU spine = Video AI + Recsys + Analytics + Safety**.
The rest of the stack is **Cursor-driven, cloud-native, and scale-ready**.

Would you like me to also **draft the matching org structure** (Tiny Team style: 4–6 ultra-specialist hires around this GPU spine) so you know who to bring in first?"